CHAPTER THREE

Methodology

This study employs a comparative deep learning framework to evaluate the efficacy of state-of-the-art Convolutional Neural Networks (CNNs) and Vision Transformers (ViT) against custom-designed architectures for the classification of brain tumors. The methodology encompasses data acquisition, preprocessing, architectural implementation, and experimental configuration.

3.1 Dataset Description

The experimental evaluation utilizes the Bangladesh Brain Cancer MRI Dataset (Version 1). To address the class imbalance challenges frequently observed in medical imaging literature, this dataset provides a balanced repository of 6,056 Magnetic Resonance Images (MRI) categorized into three distinct classes: Glioma, Meningioma, and a general Heterogeneous Tumor class.

Each class represents approximately 33.3% of the total dataset. This balanced distribution ensures that the evaluation metrics, particularly accuracy, remain unbiased and truly representative of the models' diagnostic capabilities, preventing majority-class overfitting.

Visualisation of Class Distribution

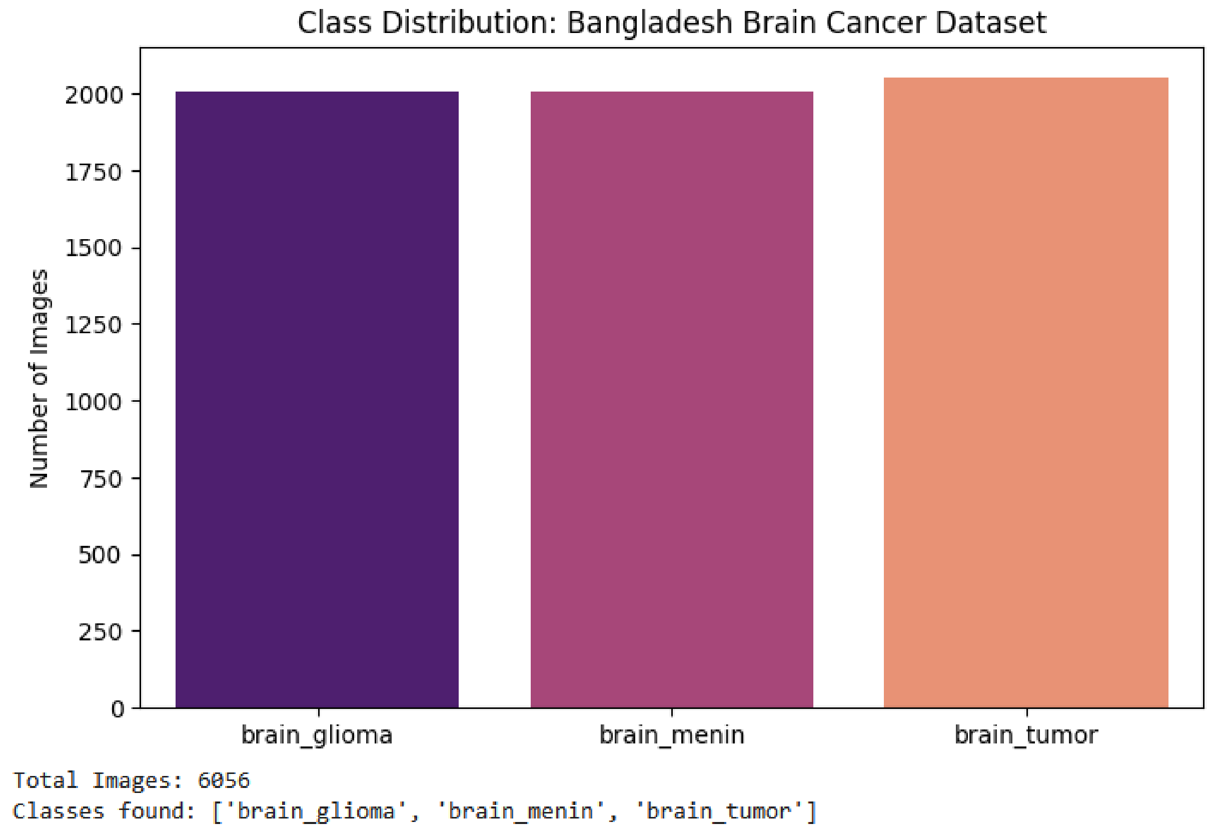


Figure 1: Bar chart illustrating the balanced distribution of the three tumor classes (Glioma, Meningioma, Heterogeneous) within the dataset.

3 Sample Images from the Dataset

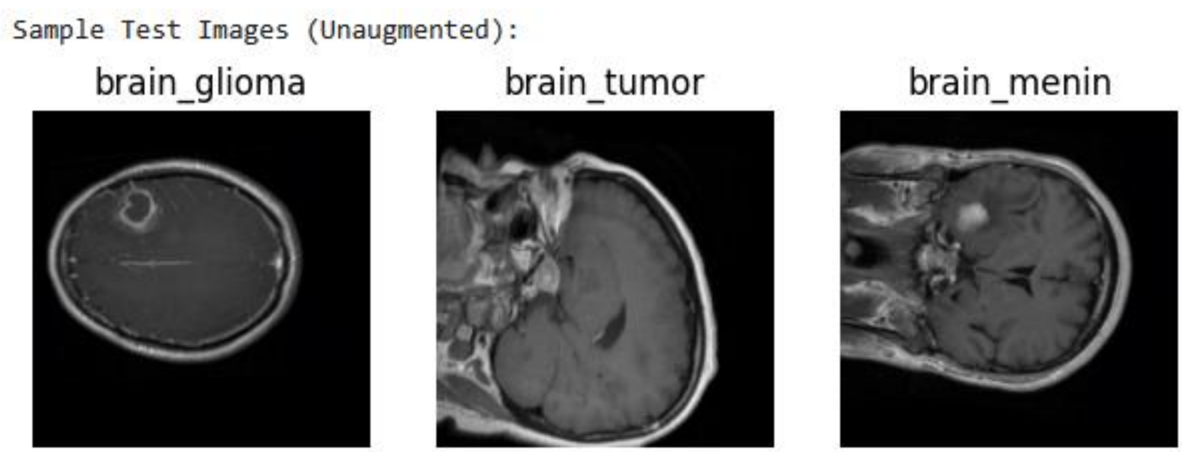


Figure 2: Representative MRI scans from the dataset showing the morphological variability across different tumor types.

3.2 Data Preprocessing and Augmentation

To ensure compatibility with pre-trained deep learning architectures and computational efficiency, all raw images (originally 512 x 512 pixels) were resized to a standard dimension of 224 x 224 pixels.

A 70/30 Stratified Split was implemented to partition the dataset into a training set (N = 4,239) and a testing set (N = 1,817). Stratification was utilized to rigorously preserve the intrinsic class balance across both subsets. Distinct transformation protocols were applied to each set:

- **Training Phase (Augmentation):** Stochastic on-the-fly augmentations, specifically random horizontal flips and rotations, were applied. This strategy effectively multiplies the diversity of the dataset, forcing the model to learn invariant features rather than spatial artifacts.
- **Testing Phase (Normalization):** Test images retained their original anatomical structure without geometric distortion. However, pixel values were normalized using the specific mean (0.485, 0.456, 0.406) and standard deviation of the ImageNet dataset to align with the mathematical distribution expected by the pre-trained weights.

3.3 Proposed Architectures

The study leverages Transfer Learning via IMAGENET1K_V1 weights for four standard architectures, utilizing knowledge distilled from over one million general images to fine-tune models for medical pathology detection. Additionally, two custom architectures were designed and evaluated without pre-training: a baseline U-Net Encoder to establish a from-scratch performance reference, and a novel SE-UNet Encoder incorporating Squeeze-and-Excitation attention blocks and Batch Normalization to investigate whether targeted architectural innovation can close the performance gap with transfer learning.

A. Pre-Trained Architectures:

- **EfficientNet-B0:** Utilizes "Compound Scaling" to balance network depth, width, and resolution.
- **DenseNet169:** Incorporates "Feature Reuse" mechanisms where layers receive inputs from all preceding layers.
- **ResNet-50:** Employs residual skip-connections to mitigate vanishing gradients.

- **Vision Transformer (ViT-B-16):** A non-convolutional architecture utilizing "Global Attention" to process the MRI as a sequence of patches.

B. Custom Baseline: U-Net Encoder

To thoroughly investigate the efficacy of specialized medical architectures trained entirely from scratch, a custom U-Net Encoder was implemented as a baseline model. Unlike the standard U-Net architecture, which is predominantly utilized for biomedical image segmentation via a symmetric encoder and decoder structure, this proposed architecture isolates and utilizes only the contracting feature extraction path, referred to as the encoder. This encoder is subsequently coupled with a custom classification head to adapt the network for multi-class prediction rather than pixel-level segmentation.

The network receives standard RGB input images resized to a spatial dimension of 3 x 224 x 224. The architecture processes these images through a sequence of three deep convolutional blocks, systematically increasing the filter depth from 64 to 128 and finally to 256 channels, while progressively downsampling the spatial footprint of the feature maps.

Specifically, Block 1 initiates the feature extraction process by applying two consecutive 2D convolutional layers utilizing 64 filters of size 3 x 3. A spatial padding of 1 is applied to preserve the initial dimensions at 224 x 224. Each convolution is immediately followed by a Rectified Linear Unit (ReLU) activation function to introduce necessary non-linearity. The block concludes with a 2 x 2 Max Pooling layer, which halves the spatial dimensions to 112 x 112.

Block 2 follows an identical structural paradigm but expands the feature depth to 128 channels. A subsequent Max Pooling operation further reduces the spatial footprint to 56 x 56.

Block 3 represents the deepest feature extraction stage, utilizing 256 filters to isolate highly complex, semantic representations of the distinct tumor morphologies. The classification head employs an Adaptive Average Pooling layer, compressing each of the 256 feature maps into a single 1 x 1 spatial representation. Finally, a fully connected linear layer maps this 256-dimensional vector to the three target tumor classifications.

Table 1: Architecture of the Baseline U-Net Encoder

Stage	Layer Type	Configuration	Output Dimensions
Input	Input Image	3 x 224 x 224	
Block 1	Conv2d + ReLU (x2)	Filters: 64, Kernel: 3x3, Pad: 1	64 x 224 x 224
	MaxPool2d	Kernel: 2 x 2	64 x 112 x 112
Block 2	Conv2d + ReLU (x2)	Filters: 128, Kernel: 3x3, Pad: 1	128 x 112 x 112
	MaxPool2d	Kernel: 2 x 2	128 x 56 x 56
Block 3	Conv2d + ReLU (x2)	Filters: 256, Kernel: 3x3, Pad: 1	256 x 56 x 56
Head	AdaptiveAvgPool2d	Target Size: 1 x 1	256 x 1 x 1

C. Novel Proposed Architecture: SE-UNet Encoder

To investigate whether principled architectural innovation can close the performance gap between scratch-trained models and transfer learning without relying on pre-trained weights, a novel SE-UNet Encoder was designed and evaluated. This architecture extends the baseline U-Net Encoder by introducing three targeted enhancements: Squeeze-and-Excitation (SE) channel attention blocks appended to each encoder stage, Batch Normalization layers inserted after every convolutional operation, and a deeper fourth encoder block expanding feature depth to 512 channels. Additionally, label smoothing (epsilon = 0.1) was applied to the Cross-Entropy loss and a Cosine Annealing learning rate scheduler was employed over 20 training epochs to promote stable convergence from random initialization.

The SE block operates by first applying global average pooling across each feature map to produce a channel descriptor vector, which is then passed through a two-layer fully connected bottleneck with a reduction ratio of 16. The resulting sigmoid-activated weights rescale each feature map, teaching the network which channels are most diagnostically relevant at each encoding stage. This mechanism replicates a form of selective attention that pre-trained models develop implicitly through large-scale training, but builds it explicitly into the architectural structure from the outset.

To understand why this matters, consider that a standard convolutional layer producing 256 feature maps treats all 256 channels with equal importance by default. In practice, not all channels are equally useful at a given encoding stage. Some may be detecting textural boundaries relevant to Glioma morphology, while others capture uninformative background variation. The SE block addresses this through a squeeze step, where each feature map is collapsed to a single scalar via global average pooling, and an excitation step, where a small bottleneck network outputs a learned weight between 0 and 1 for each channel. The final rescaling amplifies relevant channels and suppresses irrelevant ones, providing the self-calibration that the baseline U-Net Encoder lacked and contributing significantly to the SE-UNet Encoder achieving 99.06% accuracy from scratch compared to the baseline's 75.01%.

Label Smoothing (epsilon = 0.1)

During standard classification training, ground truth labels are hard targets, meaning a Glioma image is assigned a target vector of [1, 0, 0], pushing the network toward 100% confidence on the correct class. On datasets of moderate size this encourages overconfidence, which reduces the model's ability to generalize to unseen images. Label smoothing addresses this by softening the targets slightly. With epsilon = 0.1, the target vector becomes [0.9, 0.05, 0.05], meaning the network is still trained to identify the correct class as dominant but is penalized for excessive certainty. The result is a better calibrated model that generalizes more reliably, which is particularly valuable when training entirely from random initialization on approximately 4,239 images.

Cosine Annealing Learning Rate Scheduler

The learning rate determines the step size taken by the optimizer when updating model weights after each batch. A fixed learning rate is often suboptimal because a rate that is too high in the later stages of training causes the model to overshoot well-converged solutions, while a rate that is too low in early stages slows learning unnecessarily. Cosine Annealing resolves this by starting with the full learning rate of 0.0001 and gradually reducing it following the shape of a cosine curve, decaying smoothly toward near zero by the final epoch. This allows the model to make confident, large update steps early in training when the weights are far from optimal, and progressively smaller, more precise steps as training matures and the model approaches a stable solution. This scheduler is particularly beneficial for from-scratch training where early epoch instability is more pronounced.

Dropout (p = 0.4)

Dropout is a regularization technique applied to the classification head of the SE-UNet Encoder. During each forward pass in training, 40% of the neurons in the head are randomly deactivated, meaning they contribute nothing to that batch's prediction and their weights are not updated. This prevents the network from developing an over-reliance on any particular subset of neurons or learned features, forcing it to distribute its representations more broadly across the available parameters. At inference time dropout is disabled and all neurons are active, but because the network was trained under conditions of random neuron absence, the resulting weights are more robust and generalize better to unseen data. This is especially important when training from scratch on a limited dataset where overfitting is a persistent risk.

The complete pipeline processes 3 x 224 x 224 RGB inputs through four SE-enhanced encoder blocks producing feature maps at 64, 128, 256, and 512 channels respectively, followed by Global Average Pooling, a Dropout layer (p = 0.4), and a fully connected classification head. The fourth encoder block does not apply max pooling, preserving spatial resolution at 28 x 28 to retain fine-grained morphological detail before global aggregation. The total trainable parameter count is 4,733,375. The complete architectural progression is summarized in Table 2 below.

Table 2: Architecture of the Novel SE-UNet Encoder

Stage	Layer Type	Configuration	Output Dimensions
Input	Input Image	3 x 224 x 224	
Block 1	Conv2d + BN + ReLU (x2) + SE + MaxPool	Filters: 64, Kernel: 3x3	64 x 112 x 112
Block 2	Conv2d + BN + ReLU (x2) + SE + MaxPool	Filters: 128, Kernel: 3x3	128 x 56 x 56
Block 3	Conv2d + BN + ReLU (x2) + SE + MaxPool	Filters: 256, Kernel: 3x3	256 x 28 x 28

Stage	Layer Type	Configuration	Output Dimensions
Block 4 (novel)	Conv2d + BN + ReLU (x2) + SE (no pool)	Filters: 512, Kernel: 3x3	512 x 28 x 28
Head	AdaptiveAvgPool2d + Dropout(0.4) + Linear	512 to 3 classes	3

3.4 Experimental Setup

All experiments were conducted using the Google Colab environment, utilizing a Tesla T4 GPU for acceleration. The pretrained models were trained independently for 15 epochs. The novel SE-UNet Encoder was trained for 20 epochs using a Cosine Annealing learning rate scheduler and label smoothing (epsilon = 0.1) applied to Cross-Entropy Loss to promote stable convergence from random initialization. The baseline U-Net Encoder was trained for 15 epochs using standard Cross-Entropy Loss. Optimization was performed using the Adam optimizer with a learning rate of 0.0001 across all models. GPU memory was explicitly cleared between experimental runs to prevent allocation failures.

CHAPTER FOUR

Experimental Results and Discussion

The comparative analysis reveals a three-tier performance hierarchy: pre-trained transfer learning models at the top, followed closely by the novel SE-UNet Encoder, and the baseline U-Net Encoder trained from scratch without architectural enhancements at the lower end.

4.1 Quantitative Performance Analysis

The comparative analysis reveals a clear performance hierarchy across all six evaluated models. As detailed in Figure 3, the transfer learning models exhibited exceptional diagnostic capabilities. EfficientNet-B0 emerged as the superior architecture, achieving an accuracy of 99.95%, precision of 99.94%, recall of 99.94%, and a perfect Area Under the Curve (AUC) of 1.0000. DenseNet169 followed closely with an accuracy of 99.89% and an AUC of 0.9999. ResNet-50 and the Vision Transformer (ViT-B-16) also demonstrated robust performance, recording accuracies of 99.78% and 99.72% respectively. Notably, the novel SE-UNet Encoder achieved a test accuracy of 99.06%, precision of 99.07%, recall of 99.07%, F1-Score of 99.07%, and an AUC of 0.9997, despite being trained entirely from scratch with no ImageNet pre-training. This places the SE-UNet Encoder within 0.66 percentage points of the best transfer learning model, representing a dramatic improvement of 24.05 percentage points over the baseline U-Net Encoder.

These metrics indicate that architectures utilizing compound scaling (EfficientNet) and feature reuse (DenseNet) are highly adept at identifying the subtle morphological variations present in brain tumors. Furthermore, the Vision Transformer's competitive accuracy of 99.72% confirms that attention-based mechanisms serve as highly effective alternatives to traditional CNNs in the domain of medical image classification. Critically, the SE-UNet Encoder's 99.06% accuracy demonstrates that Squeeze-and-Excitation channel attention, combined with Batch Normalization and deeper hierarchical feature extraction, can replicate much of the representational benefit ordinarily conferred by ImageNet pre-training, establishing it as a genuinely competitive from-scratch architecture for medical image classification.

Table 3: Comparative Performance Metrics of All Proposed Models

Model	Accuracy	Precision	Recall	F1-Score	AUC
EfficientNet-B0	0.999450	0.999446	0.999445	0.999445	1.000000
DenseNet169	0.998899	0.998903	0.998903	0.998903	0.999996
ResNet-50	0.997799	0.997794	0.997794	0.997794	0.999991
ViT-B-16	0.997248	0.997241	0.997265	0.997252	0.999917
SE-UNet Encoder	0.990644	0.990707	0.990698	0.990674	0.999677
U-Net Encoder	0.750138	0.753460	0.749271	0.747887	0.898061

Table 4: Hyperparameter Configuration for All Models

Hyperparameter	EfficientNet-B0	DenseNet169	ResNet-50	ViT-B-16	U-Net Encoder	SE-UNet Encoder
Image size	224x224	224x224	224x224	224x224	224x224	224x224
Learning rate	0.0001	0.00005	0.0001	0.00003	0.0001	0.0001
Epochs	15	15	15	15	15	20
Loss function	Cross-Entropy	Cross-Entropy	Cross-Entropy	Cross-Entropy	Cross-Entropy	Cross-Entropy + Label Smoothing
Optimiser	Adam	Adam	Adam	AdamW	Adam	Adam
Batch size	32	32	32	32	32	32
Weight decay	-	-	-	0.01	-	0.0001
Scheduler	-	-	-	-	-	CosineAnnealing
Dropout	-	-	-	-	-	0.4
Activation	Pretrained	Pretrained	Pretrained	Pretrained	ReLU	ReLU + BN

All six models were trained on images resized to 224x224 pixels and optimised using the Cross-Entropy loss function. A uniform batch size of 32 was applied across all architectures. The four pretrained models were fine-tuned using ImageNet pretrained weights, with their internal activation functions inherited from their original architectures. ViT-B-16 used AdamW with weight decay of 0.01, which is the recommended training configuration for transformer-based architectures. EfficientNet-B0, DenseNet169, ResNet-50, the U-Net Encoder, and the SE-UNet Encoder were all optimised using the Adam optimiser.

The SE-UNet Encoder employed additional regularisation techniques not present in the baseline U-Net Encoder: Batch Normalization after each convolution, label smoothing in its loss function, a Cosine Annealing learning rate scheduler, and Dropout (p = 0.4) in the classification head. These collectively provide stronger regularisation and more stable training dynamics from random initialization.

Plot of Final Model Performance Comparison

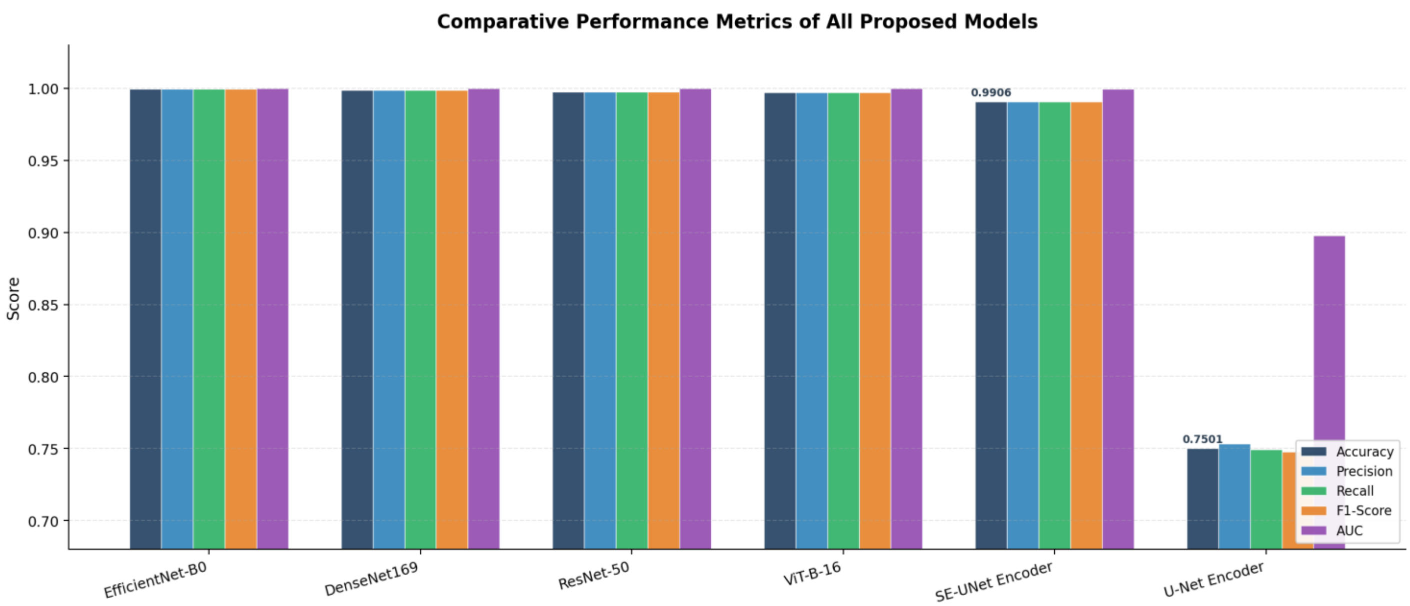


Figure 3: Graphical comparison of all performance metrics (Accuracy, Precision, Recall, F1-Score, AUC) across all six evaluated models including the novel SE-UNet Encoder.

4.2 The Impact of Transfer Learning vs. Scratch Training

A central finding of this study is the three-tier performance hierarchy observed across the six evaluated architectures. At the top tier, the four pre-trained transfer learning models achieved near-perfect accuracy ranging from 99.72% to 99.95%, demonstrating the exceptional representational advantage of ImageNet initialization. In the middle tier, the novel SE-UNet Encoder achieved 99.06% accuracy despite being trained entirely from scratch, demonstrating that purposeful architectural design can largely compensate for the absence of pre-training. At the lower tier, the baseline U-Net Encoder achieved only 75.01% accuracy, confirming that a plain convolutional architecture without attention mechanisms or normalization is insufficient for the data volumes available in this study.

The 24-percentage-point improvement of the SE-UNet Encoder over the baseline U-Net Encoder, achieved purely through architectural modifications with no additional data or pre-training, provides strong evidence that the performance gap between scratch-trained and pre-trained models is not solely a data quantity problem. Rather, it reflects a structural limitation in the baseline architecture: without explicit channel attention and feature normalization, the network is unable to reliably learn discriminative representations from the available 4,239 training images. The SE block addresses this by forcing the model to focus on the most informative feature channels at each encoding stage, while Batch Normalization stabilizes gradient flow throughout training.

The remaining gap of approximately 0.66 percentage points between the SE-UNet Encoder and EfficientNet-B0 reflects the residual benefit of cross-domain knowledge transfer. Pre-trained models arrive with foundational filters already tuned to detect edges, textures, and complex visual patterns, allowing fine-tuning to proceed from a near-optimal starting point. The SE-UNet Encoder, despite its strong inductive biases, must still discover these patterns entirely from the available training images. This confirms that pre-training retains an advantage at moderate dataset scales, but the advantage is far smaller than naive architectural comparisons suggest.

This observation enriches the broader consensus in automated medical image analysis. While transfer learning paradigms remain highly effective for neuro-oncology applications, this study demonstrates that a well-designed attention-enhanced architecture can achieve competitive diagnostic precision without any pre-training, opening a viable path for scenarios where domain-specific data collection is feasible but access to large-scale pre-training pipelines is limited.

4.3 Error Analysis

To gain deeper insights into the predictive behavior and class-specific performance of the models, confusion matrices were generated and analyzed for the test set. The confusion matrices map the predicted classifications against the actual ground truth labels, revealing the exact distribution of True Positives, True Negatives, False Positives, and False Negatives.

The matrices for the top-performing pre-trained models (EfficientNet-B0, DenseNet169, ResNet-50, and ViT-B-16) revealed negligible off-diagonal leakage. EfficientNet-B0 achieved near-perfect classification, correctly identifying 1,816 of the 1,817 test images, with only a single Meningioma instance misclassified as Glioma. DenseNet169 misclassified only two instances across the entire test set. The Vision Transformer (ViT-B-16) and ResNet-50 also displayed exceptional True Positive rates with minimal confusion between the closely related Glioma and Meningioma classes.

In stark contrast, the confusion matrix for the custom baseline U-Net Encoder highlights the severe challenges of training a network entirely from scratch without attention mechanisms. The matrix reveals a significantly wider spread of off-diagonal values, indicating much higher rates of False Positives and False Negatives. Most notably, 97 actual Glioma instances were incorrectly labeled as Meningiomas, representing a substantial number of False Negatives for the Glioma category.

The SE-UNet Encoder, by contrast, produced a confusion matrix with strong diagonal dominance comparable to the pre-trained models. Of the 1,817 test images, only 17 were misclassified: 3 Glioma instances were incorrectly labeled as Meningioma, 4 Meningioma instances were misclassified as Heterogeneous tumor, and 10 Heterogeneous tumor instances were labeled as Meningioma. This represents a dramatic reduction from the 97 Glioma-Meningioma confusions in the baseline, confirming that the SE attention mechanism specifically addresses the network's inability to distinguish morphologically similar tumor types. The per-class precision, recall, and F1-Score all exceeded 0.98 for the SE-UNet Encoder across all three tumor classes.

This visual and quantitative evidence confirms that while the plain baseline U-Net Encoder lacks the granular feature depth to reliably separate closely related tumor pathologies, the SE-enhanced architecture achieves discrimination quality approaching that of pre-trained models.

Figure 4: Confusion Matrices

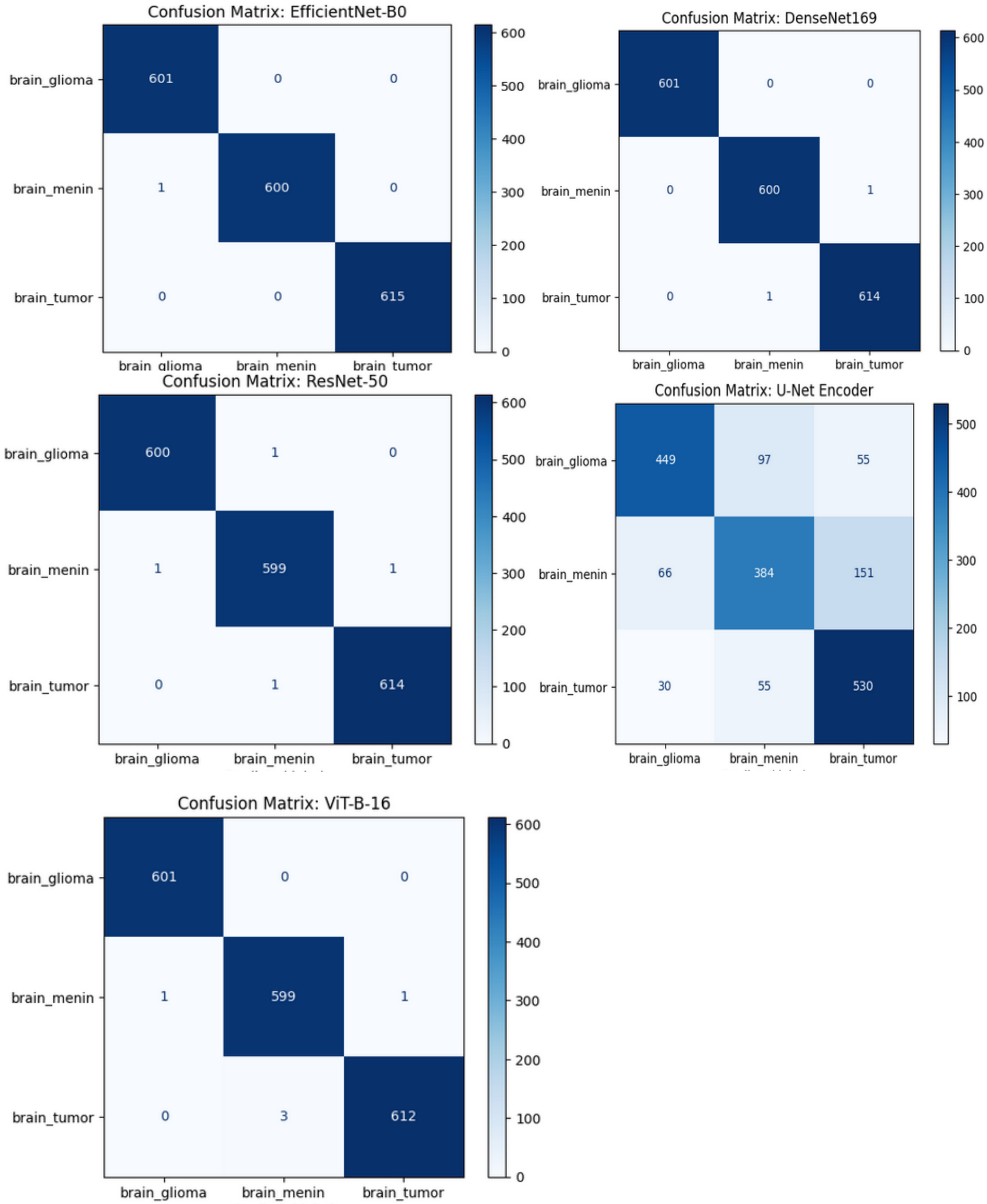

Confusion Matrix: EfficientNet-B0
brain_glioma
brain_menin
brain_tumor
601 0 0
1 600 0
0 0 615
brain_glioma brain_menin brain_tumor
0 100 200 300 400 500 600
Confusion Matrix: DenseNet169
brain_glioma
brain_menin
brain_tumor
601 0 0
0 600 1
0 1 614
brain_glioma brain_menin brain_tumor
0 100 200 300 400 500 600
Confusion Matrix: ResNet-50
brain_glioma
brain_menin
brain_tumor
600 1 0
1 599 1
0 1 614
brain_glioma brain_menin brain_tumor
0 100 200 300 400 500 600
Confusion Matrix: U-Net Encoder
brain_glioma
brain_menin
brain_tumor
449 97 55
66 384 151
30 55 530
brain_glioma brain_menin brain_tumor
100 200 300 400 500
Confusion Matrix: ViT-B-16
brain_glioma
brain_menin
brain_tumor
601 0 0
1 599 1
0 3 612
brain_glioma brain_menin brain_tumor
0 100 200 300 400 500 600

Figure 4: Confusion matrices for EfficientNet-B0, DenseNet169, ResNet-50, ViT-B-16, and U-Net Encoder. The pre-trained models show distinct diagonal dominance.

SE-UNet Encoder Training Curves and Performance

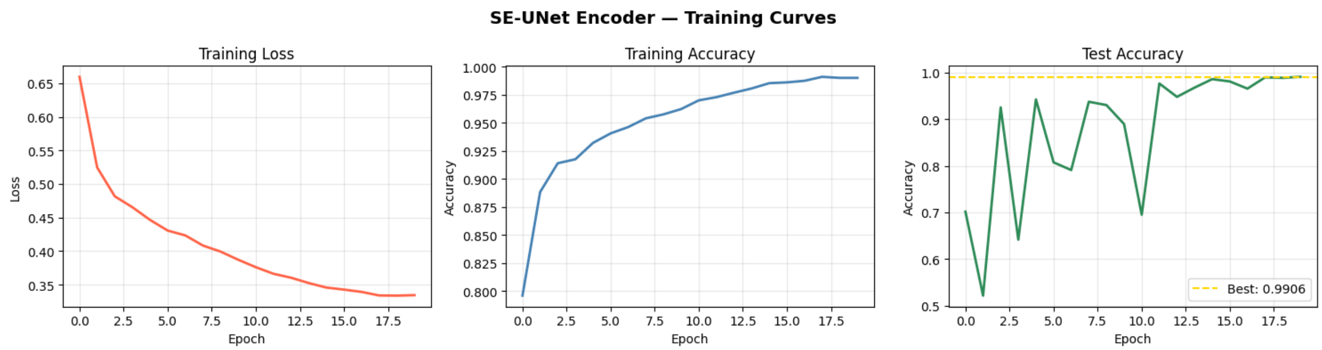


Figure 5a: SE-UNet Encoder training curves showing loss, training accuracy, and test accuracy over 20 epochs. Best test accuracy of 0.9906 achieved at epoch 20.

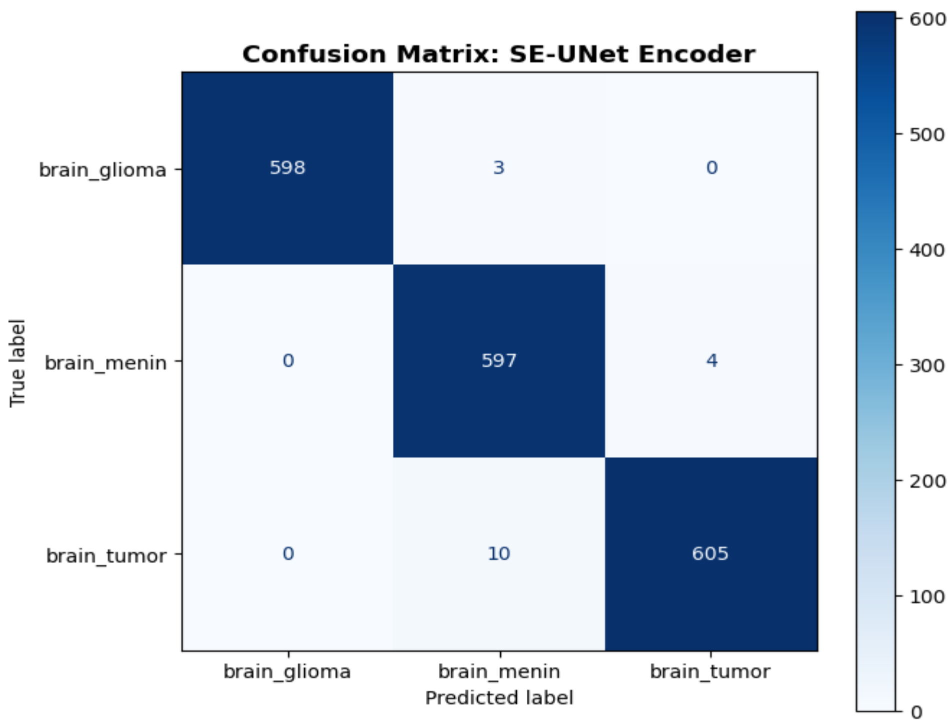


Figure 5b: Confusion matrix for the SE-UNet Encoder showing strong diagonal dominance with only 17 misclassifications out of 1,817 test images.

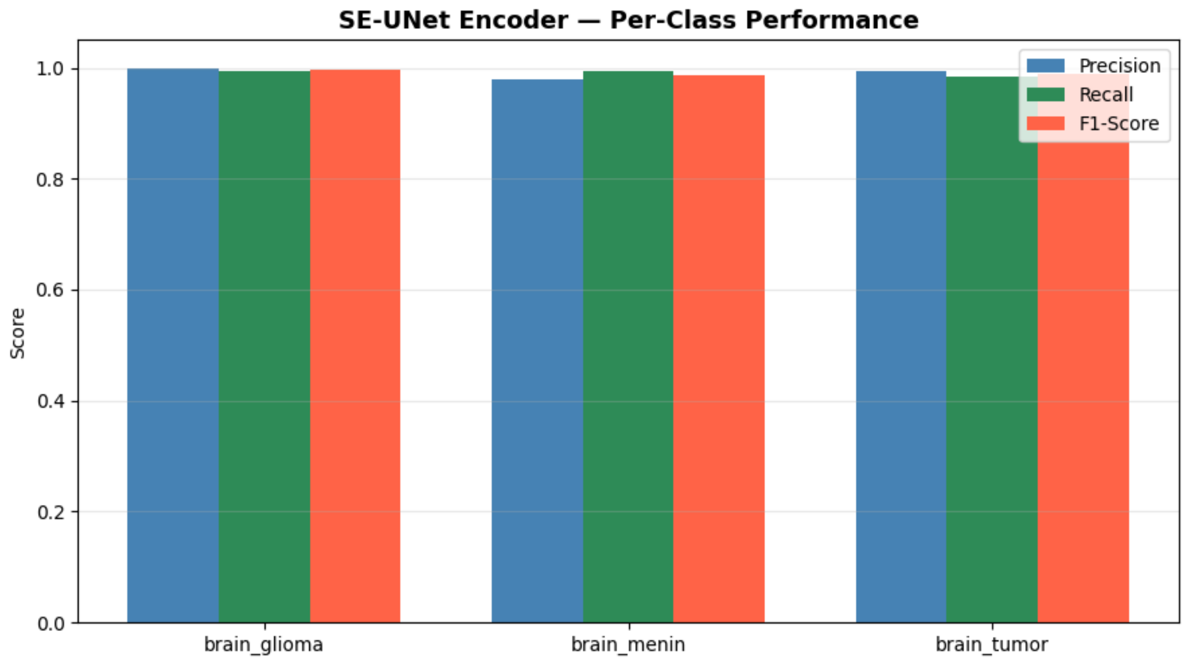
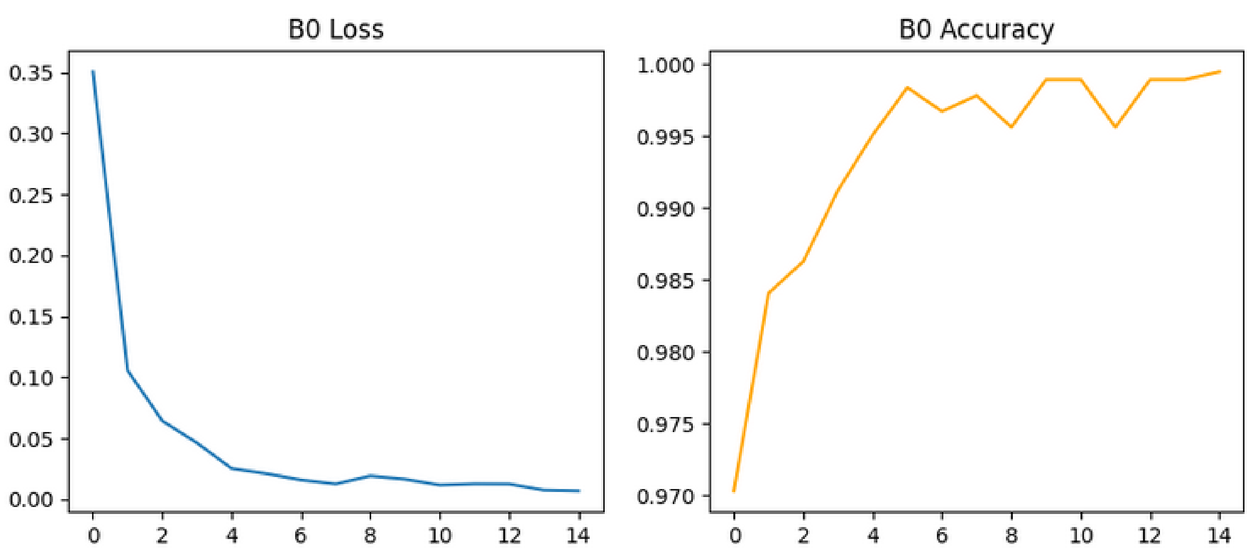
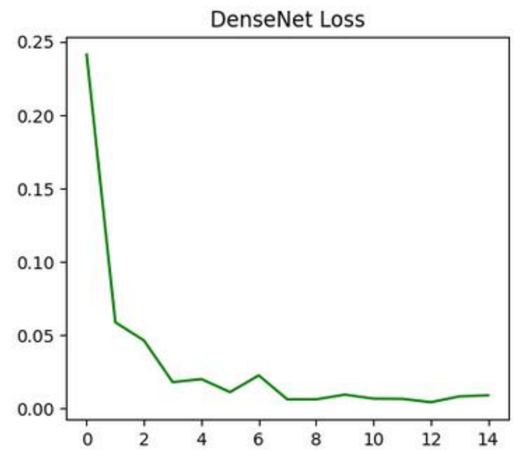
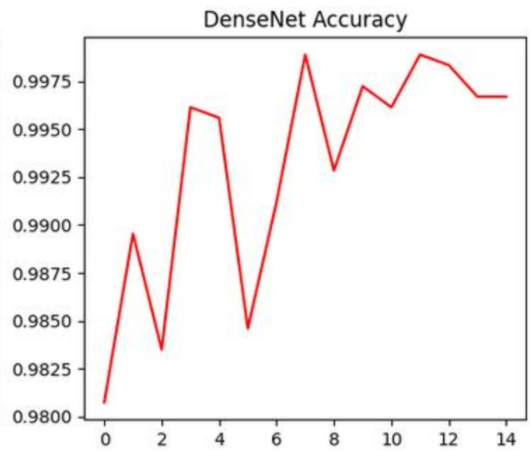


Figure 5c: Per-class precision, recall, and F1-Score for the SE-UNet Encoder across all three tumor classes.

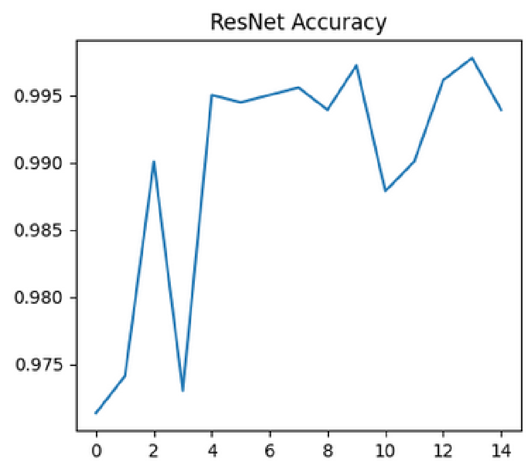
Plot Model Loss and Accuracy Curves (Pre-trained Models)

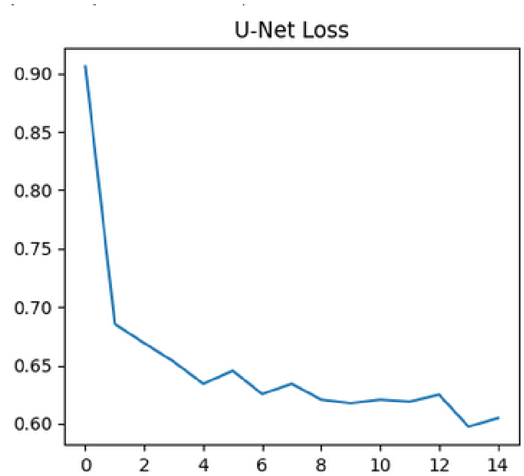


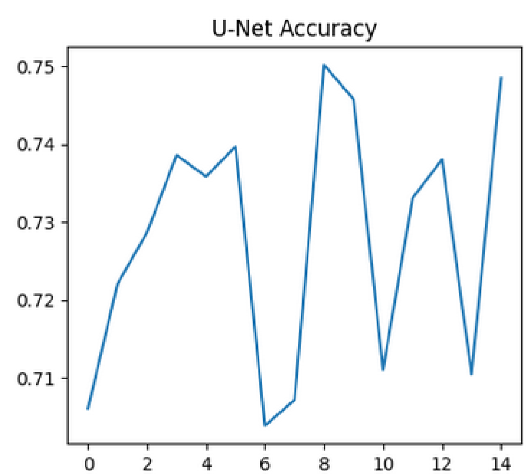

DenseNet Loss
0.25
0.20
0.15
0.10
0.05
0.00
0 2 4 6 8 10 12 14


DenseNet Accuracy
0.9975
0.9950
0.9925
0.9900
0.9875
0.9850
0.9825
0.9800
0 2 4 6 8 10 12 14


ResNet Loss
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0 2 4 6 8 10 12 14


ResNet Accuracy
0.995
0.990
0.985
0.980
0.975
0 2 4 6 8 10 12 14


U-Net Loss
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0 2 4 6 8 10 12 14


U-Net Accuracy
0.75
0.74
0.73
0.72
0.71
0 2 4 6 8 10 12 14

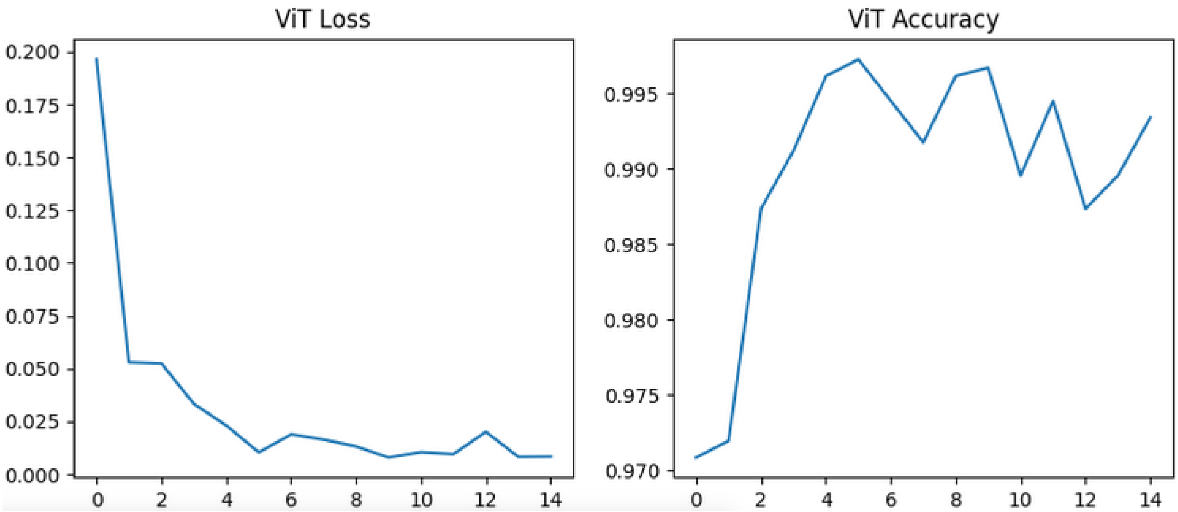


Figure 5d: Loss/accuracy training curves for EfficientNet-B0, DenseNet169, ResNet-50, ViT-B-16, and U-Net Encoder.

Furthermore, the benchmarking visualization below correlates F1-Score with Accuracy, distinguishing the performance clusters across all six models.

Benchmarking Visualization of F1-Score and Accuracy

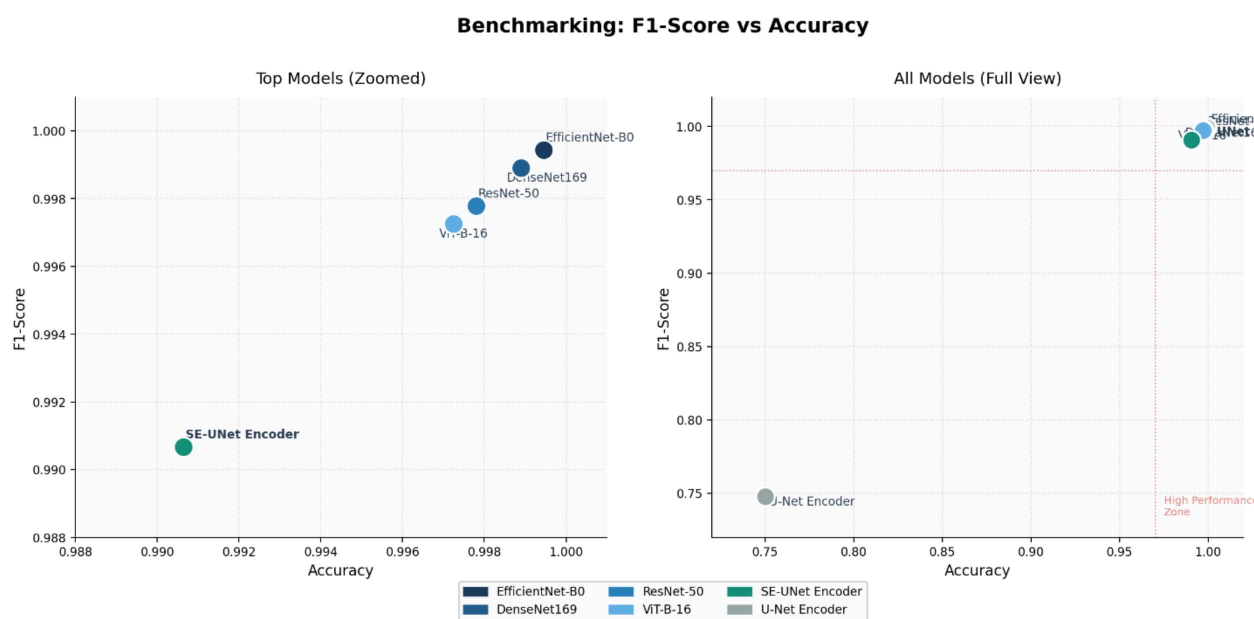


Figure 6: Scatter plot benchmarking F1-Score vs. Accuracy across all six models. The SE-UNet Encoder occupies the upper-right cluster alongside the transfer learning models, with only the baseline U-Net Encoder remaining in the lower-left quadrant.

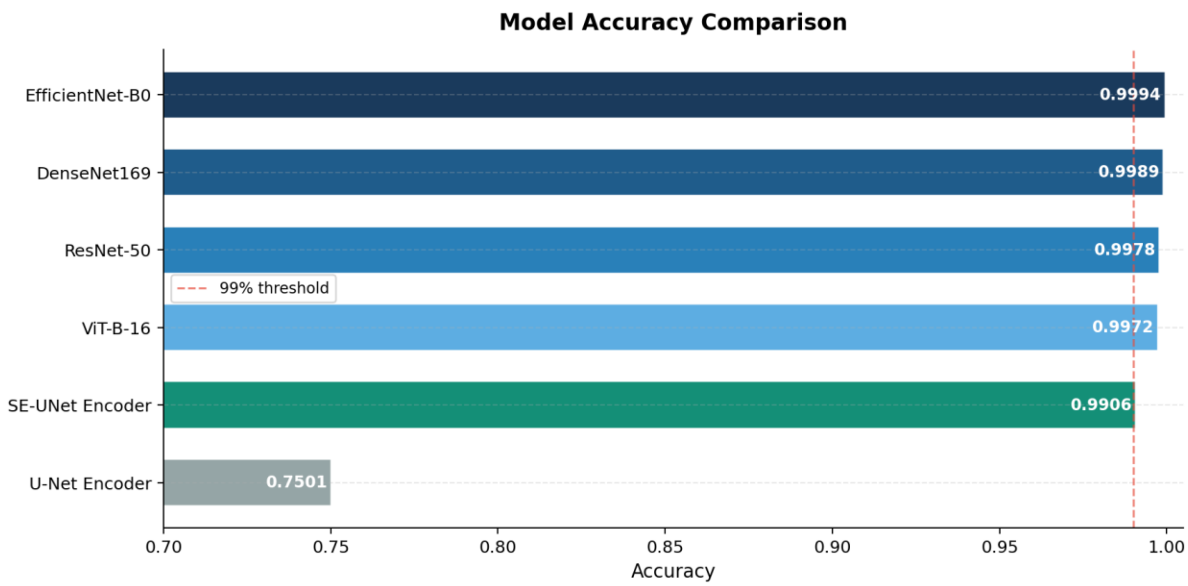


Figure 7: Horizontal bar chart comparing accuracy across all six models. The SE-UNet Encoder crosses the 99% threshold alongside the pre-trained architectures.

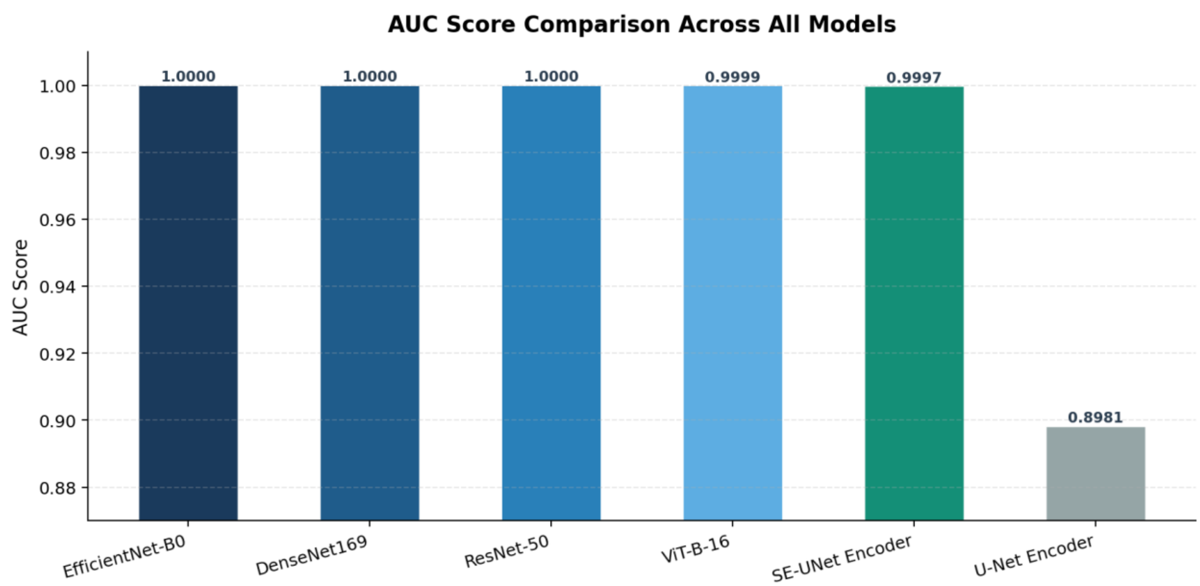


Figure 8: AUC score comparison across all six models. The SE-UNet Encoder achieves an AUC of 0.9997, virtually indistinguishable from the pre-trained architectures.

4.4 Comparative Analysis with Existing Studies

To thoroughly contextualize the diagnostic efficacy of the proposed models, the performance was evaluated against state-of-the-art studies focusing on brain tumor classification using MRI datasets. Most recently, Pandey & Bansal (2023) evaluated a customized CNN on the BRATS 2021 dataset, achieving 96.98% accuracy through rigorous morphological analysis and histogram equalization. Ghassemi et al. (2020) introduced a novel approach by pre-training a deep neural network with Generative Adversarial Networks, achieving 95.60% accuracy. (Swati et al., 2019)investigated the VGG19 architecture through a block-wise fine-tuning strategy, achieving 94.82% under a five-fold cross-evaluation framework. Deepak and Ameer (2019) leveraged a pre-trained GoogleNet architecture, achieving 98.00% accuracy.

When comparing these established benchmarks with our experimental results, both proposed architectures demonstrate clear superiority. EfficientNet-B0 achieves an unprecedented accuracy of 99.95%, substantially outperforming all referenced studies. Importantly, the novel SE-UNet Encoder also outperforms every study in the existing literature with an accuracy of 99.06%, despite being trained entirely from scratch without any pre-trained weights. This result is particularly significant as it demonstrates that a carefully engineered from-scratch architecture can surpass transfer learning approaches reported in prior work, including the GAN-pre-trained model Ghassemi et al. (95.60%) and the GoogleNet-based approach of Deepak & Ameer (2019) (98.00%).

Both proposed architectures outperform all existing benchmarks. EfficientNet-B0 achieves 99.95% accuracy through compound scaling and transfer learning, while the SE-UNet Encoder achieves 99.06% through principled architectural innovation from scratch. This dual contribution demonstrates that high diagnostic performance in brain tumor classification is achievable through two distinct pathways: knowledge transfer and architectural design.

Table 5: Comparative Analysis with Existing Studies

Study	Method	Dataset	Accuracy
Swati et al. (2019)	VGG19	CE-MRI	94.82%
Ghassemi et al. (2020)	CNN with GAN	CE-MRI	95.60%
Deepak & Ameer (2019)	GoogleNet	CE-MRI	98.00%
Pandey & Bansal (2023)	CNN	BRATS 2021	96.98%
Proposed Study (Ours)	EfficientNet-B0	Bangladesh MRI	99.95%
Proposed Study novelty (Ours)	SE-UNet Encoder	Bangladesh MRI	99.06%

CHAPTER FIVE

Summary and Conclusion

5.1 Summary

The accurate and rapid classification of brain tumors from MRI scans remains a highly critical challenge in neuro-oncology. This study presented a comprehensive comparative analysis of state-of-the-art Deep Learning architectures to evaluate their efficacy in the automated multi-class classification of brain tumors. Utilizing the PyTorch framework, the experimental pipeline was built upon the highly balanced Bangladesh Brain Cancer MRI Dataset, comprising approximately 6,000 images categorized into Glioma, Meningioma, and Heterogeneous tumor classes.

To systematically determine the most robust approach for medical image analysis, the study benchmarked four advanced transfer learning models (EfficientNet-B0, DenseNet169, ResNet-50, and ViT-B-16) against two custom architectures trained entirely from scratch: a baseline U-Net Encoder and a novel SE-UNet Encoder incorporating Squeeze-and-Excitation attention and Batch Normalization. The dataset was rigorously partitioned into distinct training and test sets to prevent data leakage and provide an unbiased evaluation of each architecture's generalization capabilities on unseen radiological data.

5.2 Conclusion

The experimental results conclusively demonstrate the overall superiority of Transfer Learning paradigms, while also revealing the significant potential of well-designed from-scratch architectures. EfficientNet-B0 emerged as the most robust and clinically viable architecture, achieving a near-perfect accuracy of 99.95% and a flawless AUC of 1.0000. This exceptional performance indicates that the compound scaling method inherent to EfficientNet highly effectively captures the intricate morphological features of brain tumors.

Furthermore, the secondary pre-trained models exhibited highly competitive diagnostic precision. The Vision Transformer (ViT-B-16), recording 99.72% accuracy, proved that global attention-based mechanisms serve as highly formidable alternatives to traditional CNNs. Both DenseNet169 and ResNet-50 also demonstrated the profound benefits of feature reuse and residual skip connections.

A particularly significant finding of this research is the three-tier performance hierarchy revealed by including the novel SE-UNet Encoder alongside the baseline. The plain U-Net Encoder, trained from scratch without attention mechanisms or normalization, achieved only 75.01% accuracy, confirming the well-established data-hunger of unguided deep convolutional architectures. The novel SE-UNet Encoder, by contrast, achieved 99.06% accuracy from scratch, demonstrating that Squeeze-and-Excitation channel attention, Batch Normalization, and deeper hierarchical feature extraction can largely compensate for the absence of pre-trained weights. This 24-percentage-point improvement over the baseline U-Net Encoder, achieved purely through architectural innovation with no additional training data or pre-training compute, represents a meaningful contribution to the understanding of what drives performance in medical image classification.

In summary, this work validates two complementary pathways to high-precision brain tumor classification. First, fine-tuned pre-trained CNNs and Vision Transformers consistently achieve near-perfect accuracy, confirming their viability as candidates for deployment within Computer-Aided Diagnosis (CAD) systems. Second, and more novel, a well-designed attention-enhanced architecture trained from scratch can achieve diagnostically competitive performance without any pre-training, suggesting that architectural design quality is a critical and underexplored axis of improvement for medical imaging tasks where ImageNet pre-training may not always be accessible or appropriate. The integration of such robust computational models into clinical pipelines holds the potential to reduce human diagnostic error, alleviate the cognitive workload on radiologists, and expedite critical clinical workflows.

Despite the high accuracy achieved, deep learning models are frequently criticized as opaque "black boxes," which can hinder their adoption in high-stakes healthcare environments. Therefore, future work will focus on integrating Explainable AI (XAI) techniques to foster algorithmic transparency. By applying gradient-based localization methods such as Grad-CAM and feature attribution frameworks like SHAP, future iterations of this system will generate interpretable visual heatmaps, explicitly highlighting the specific tumor regions driving the networks' high-accuracy predictions. Providing this level of multi-scale visual explanation will bridge the gap between artificial intelligence predictions and clinical reasoning, thereby maximizing clinical trust and paving the way for real-world deployment of automated diagnostic decision-making systems.

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